

Chapter One

Energy Reserves and Potential

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CHAPTER 1: Energy Reserves and Potential

Introduction

Energy reserves are part of the energy resources that, based on technical, economic and other relevant (e.g. environmental) considerations, can be recovered and for which extraction is justified.

The exact definition of reserves depends on the kind resources in focus.

Global Classification of Energy Reserves

The classification of energy reserves is guided by the United Nations Framework Classification for Fossil Energy and Mineral Reserves and Resources (UNFC 2009).

This framework evaluates resources based on: Economic and Social Viability, Field Project Status and Feasibility, and Geological Knowledge.

- Class A: Commercially recoverable resources.
- Class B: Potentially commercially recoverable resources.
- Class C: Non-commercial and other known deposits.

Energy Reserves in India

India has a diverse range of energy resources, with significant reserves of both fossil fuels and renewable energy potential.

The country's energy mix is undergoing a transformation to reduce dependence on fossil fuels and promote the use of renewable energy sources, driven by environmental, economic, and macroeconomic consideration.

This chapter highlights the energy reserves and potential in India, focusing on both fossil fuels and renewable energy resources. It explores coal, lignite, crude oil, natural gas, and renewable energy potential, providing an

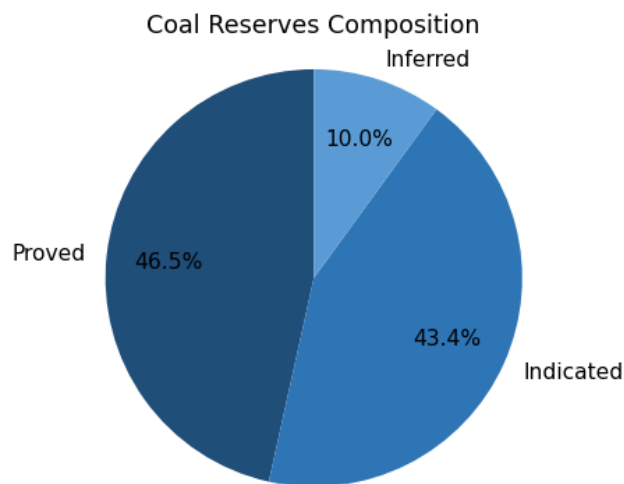


Fig 1.1 Estimated Reserves of Coal in India as on 01.04.2025

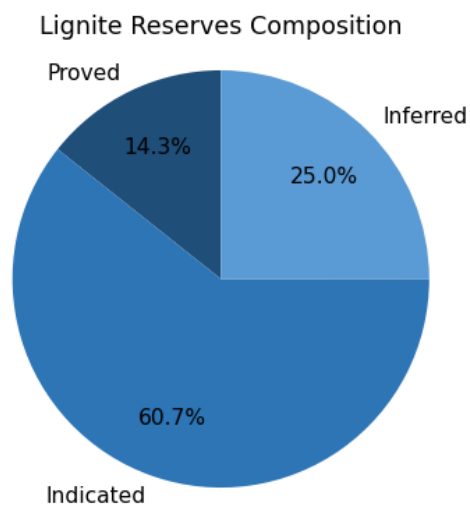


Fig 1.2 Estimated Reserves of Lignite in India as on 01.04.2025

1.1 Coal Reserves

India is home to one of the largest proven coal reserves in the world. As of 01-04-2025, the total estimated coal reserves were 400.7 billion tonnes, showing an addition of 11.29 billion tonnes compared to the previous year.

This marks a growth of 2.90% in total coal reserves during the year 2024-25(P) over the previous year (2023-24) (Table 1.1).

The distribution of coal reserves in India is concentrated in a few states like Odisha, Jharkhand and Chhattisgarh. These three states account for approximately 70% of the total coal reserves in the country. Out of the total reserves in the country, proven reserves i.e. economically viable and geologically confirmed account for approximately 55% of the total estimated coal reserves as depicted in the figure 1.1.

1.2 Lignite Reserves

1.3 Crude Oil Reserves

As of 01-04-2025, the estimated reserves of crude oil in India stood at 672.07 million tonnes, compared to 671.40 million tonnes in the previous year, reflecting a 0.10% increase over the last year (Table 1.3).

Geographically, the maximum crude oil reserves in India are concentrated in the Western Offshore region, which accounts for 31% of the total crude oil reserves. This is followed by the Assam region, which holds around 22% of the country's crude oil reserves as shown in figure 1.3.

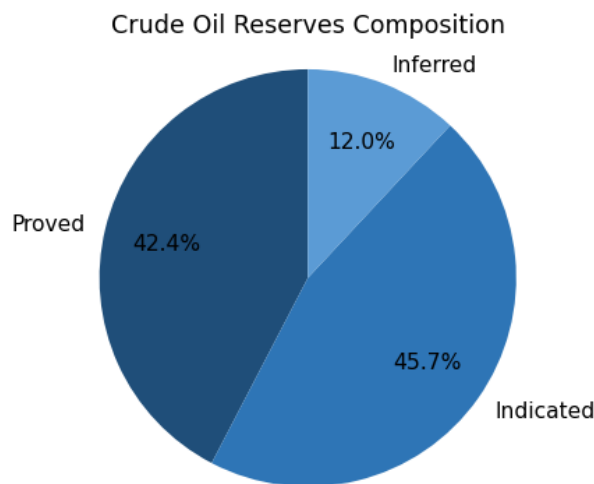


Fig 1.3: Estimated Reserves of Crude Oil in India as on 01.04.2025

1.4 Natural Gas Reserves

The estimated reserves of natural gas as of 01-04-2025 were 1,073.01 Billion Cubic Meters (BCM) compared to 1,094.19 Billion Cubic Meters (BCM) in the previous year, reflecting a decrease of 1.94% over the last year

The largest reserves of natural gas in India are located in the Western Offshore region, which holds approximately 32% of the total natural gas reserves. This is followed by the Eastern Offshore, which accounts

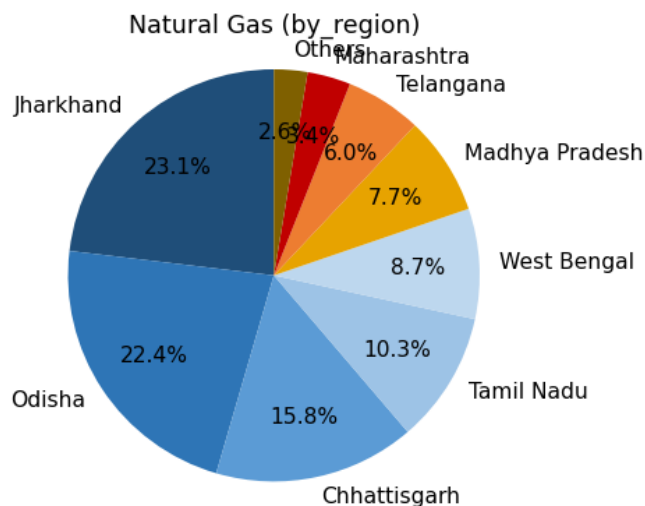


Fig 1.4: Estimated Reserves of Natural Gas in India as on 01.04.2025

1.5 Renewable Energy Potential in India

India is increasingly focusing on renewable energy sources as part of its sustainable development and mitigating impact on climate. As of March 31, 2025, the total estimated potential for renewable power generation in India

This potential comes from a variety of renewable sources, including wind, solar, biomass, small hydro, and cogeneration from bagasse (Table 1.4). Figure 1.5 shows the potential is spread across several key renewable

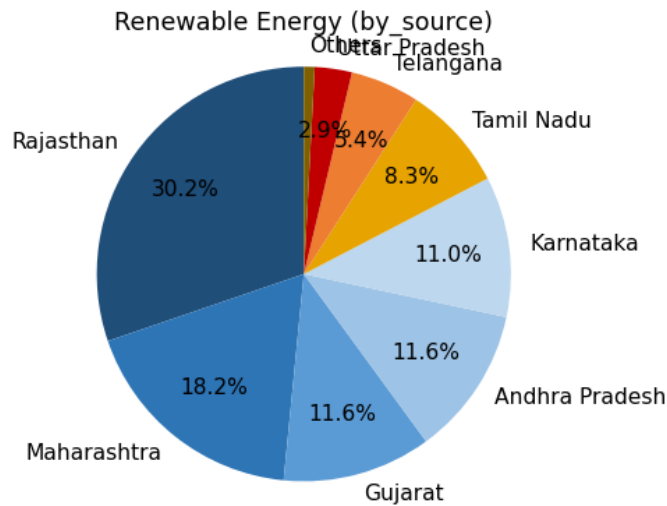


Fig 1.5: Source wise Estimated Potential of Renewable Power in India as on 31.03.2025

Solar Power: India has immense potential for solar power generation, estimated at 33,43,378 MW, which accounts for 71% of the total renewable energy potential. Solar power stands as the most significant source of

Wind Power: With an estimated potential of 1,163,856 MW (close to 25% of total renewable energy potential), it has the potential to contribute significantly in shaping the renewable energy-generation scenario of India.

Large Hydropower (Large Hydro): The potential for large hydropower generation is estimated at 133,410 MW, contributing 2.84% to the country's renewable energy potential.

Small Hydro Power (SHP): India also has a notable potential of 21,134 MW (0.45%) from small hydro power projects.

Biomass Power: Biomass, which includes agricultural waste, forest residues and other organic matter, has a potential of 28,447 MW, accounting for 0.60% of the total renewable power potential.

Cogeneration from Bagasse: India has a specific potential of 13,818 MW (0.29%) from bagasse-based cogeneration in sugar mills. This is a highly efficient form of energy generation, especially in regions with a

Geographical Distribution of Renewable Energy Potential

The geographical distribution of renewable energy potential in India (Figure 1.6) shows significant regional variations. The states with the highest renewable energy potential, as of March 31, 2025, are: Rajasthan,

Together, these five states account for around 65% of India's total renewable energy potential, highlighting their critical role in India's renewable energy future.

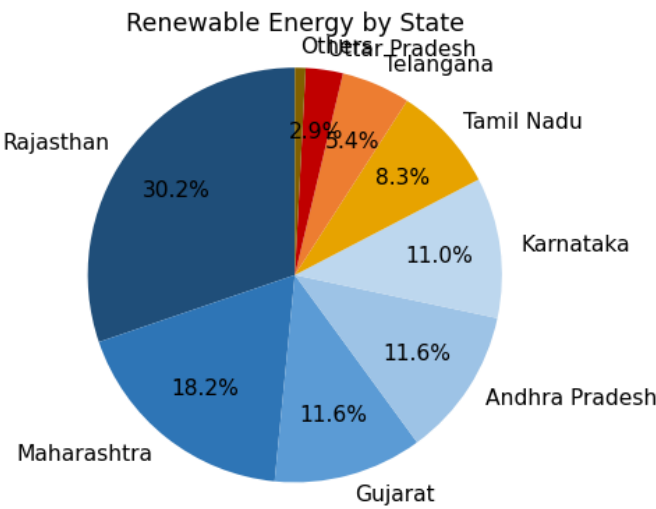


Fig 1.6: Statewise Estimated Potential of Renewable Power in India as on 31.03.2025

Table 1.1: Statewise Estimated Reserves of Coal (As on 1st April 2025)

State...	Prov...	Prov...	Indic...	Indic...	Inferr...	Inferr...	Total...	Total...	Distri...	Distri...
Jhark...	45563	45563	31439	31439	6559	6559	83561	83561	None	None
Odisha	37391	37391	34165	34165	9410	9410	80966	80966	None	None
Chha...	20460	20460	34433	34433	2228	2228	57121	57121	None	None
West...	13403	13403	13022	13022	4880	4880	31305	31305	None	None
Madh...	11958	11958	12176	12176	3848	3848	27982	27982	None	None
Telan...	10486	10486	8511	8511	2683	2683	21680	21680	None	None
Maha...	7286	7286	3060	3060	1956	1956	12302	12302	None	None
Total	146547	146547	136806	136806	31564	31564	314917	314917	None	None

Table 1.2: Statewise Estimated Reserves of Lignite(As on 1st April 2025)

States...	Prove...	Prove...	Indica...	Indica...	Inferre...	Inferre...	Total...	Total...	Distri...
Tamil...	4262	4262	24683	24683	8376	8376	37321	37321	None
Gujarat	1278	1278	283	283	1161	1161	2722	2722	None
Rajast...	1168	1168	3029	3029	2138	2138	6335	6335	None
Jamm...	0	0	20	20	7	7	27	27	None
Kerala	0	0	0	0	10	10	10	10	None
Puduc...	0	0	405	405	12	12	417	417	None
Total	6708	6708	28420	28420	11704	11704	46832	46832	None

Table 1.3: Statewise Estimated Reserves of Crude Oil and Natural Gas (As

States/UTs/Region	Crude Oil 2024	Crude Oil 2025	Natural Gas 2024	Natural Gas 2025
Andhra Pradesh	7.69	8.41	59.27	60.60
Arunachal Pradesh	2.88	0.44	7.33	7.10
Assam	145.41	144.73	164.51	163.33
Gujarat	118.86	121.20	55.46	54.74
Nagaland	2.38	2.38	0.09	0.09
Rajasthan	131.50	136.46	63.55	27.12
Tamil Nadu	8.54	9.11	36.57	38.39
Tripura	0.07	0.07	28.18	27.73
West Bengal	0.15	0.20	0.00	0.79
West Bengal (CBM)	0.00	0.00	44.37	44.93
Jharkhand (CBM)	0.00	0.00	4.86	5.67
Madhya Pradesh...	0.00	0.00	24.38	23.15
Eastern Offshore	40.67	37.25	263.34	270.67
Western Offshore	213.26	209.30	342.29	348.70
Total	671.40	672.07	1094.19	1073.01

Table 1.4: Sourcewise and Statewise Estimated Potential of Renewable

States/...	Wind P...	Small H...	Biomass...	Cogene...	Solar E...	Large...	Total	Distrib...
Jharkhand	None	None	None	None	None	None	83561	None
Odisha	None	None	None	None	None	None	80966	None
Chhatti...	None	None	None	None	None	None	57121	None
Tamil...	None	None	None	None	None	None	37321	None
West B...	None	None	None	None	None	None	31305	None
Madhy...	None	None	None	None	None	None	27982	None
Telangana	None	None	None	None	None	None	21680	None
Mahara...	None	None	None	None	None	None	12302	None
Rajasthan	None	None	None	None	None	None	6335	None
Gujarat	None	None	None	None	None	None	2722	None
Puduch...	None	None	None	None	None	None	417	None
Jammu...	None	None	None	None	None	None	27	None
Kerala	None	None	None	None	None	None	10	None
Andhra...	None	None	None	None	None	None	0	None
Karnataka	None	None	None	None	None	None	0	None
Punjab	None	None	None	None	None	None	0	None
Sikkim	None	None	None	None	None	None	0	None
Tripura	None	None	None	None	None	None	0	None
Uttar Pr...	None	None	None	None	None	None	0	None
Total	None	None	None	None	None	None	361749	None